Water Resources

The big idea

Three-fourths of the Earth is covered with water, yet only a small fraction is usable fresh water. Water is a **renewable** resource, renewed through the **hydrological cycle**, but its availability varies over space and time. Over-use, pollution and unequal access cause **water scarcity**. This chapter explores scarcity, the debate over large dams, and traditional **rainwater harvesting** methods.

Water scarcity

HOW MUCH WATER IS THERE?

96.5 per cent of the total volume of the world's water is in the **oceans**; only **2.5 per cent** is **freshwater**. Nearly **70 per cent** of this freshwater occurs as **ice sheets and glaciers** in Antarctica, Greenland and mountain regions, and a little less than **30 per cent** is stored as **groundwater**. India receives nearly **4 per cent** of the global precipitation and ranks **133rd** in the world in terms of water availability per person per annum.

Water scarcity is not only about too little water. Causes:

- **Large and growing population**: more water for domestic use and more food production.
- **Over-exploitation for irrigation**: to facilitate higher food grain production, water resources are over-exploited for dry-season agriculture; many farmers have their own wells and tube-wells, causing **falling groundwater levels**.
- **Urbanisation and industrialisation**: industries need water and power (hydroelectric). Cities with large populations and urban lifestyles have their

own groundwater pumping devices.

- **Poor quality of water**: even where there is enough water, it may be **polluted** by domestic and industrial wastes, chemicals, pesticides and fertilisers, making it hazardous for use.
- **Unequal access** among different social groups.

SCARCITY IN A WATER-RICH AREA

A region may have ample water and yet face scarcity because the water is polluted, or because it is unequally distributed. That is why conserving and managing water is a must.

Multi-purpose river projects and integrated water resources management

History of hydraulic structures in India:

- **1st century BC**: **Sringaverapura near Prayagraj (Allahabad)** had a sophisticated water harvesting system channelling the floodwater of the river Ganga.
- During **Chandragupta Maurya's** time, dams, lakes and irrigation systems were extensively built.
- Evidence of sophisticated irrigation works in **Kalinga (Odisha), Nagarjunakonda (Andhra Pradesh), Bennur (Karnataka), Kolhapur (Maharashtra)**.
- **11th century**: **Bhopal Lake**, one of the largest artificial lakes of its time.
- **14th century**: the tank in **Hauz Khas, Delhi**, was constructed by **Iltutmish** to supply water to the Siri Fort area.

DAM AND MULTI-PURPOSE PROJECT

A **dam** is a barrier across flowing water that obstructs, directs or retards the flow, often creating a reservoir, lake or impoundment. **Multi-purpose projects** integrate several uses: irrigation, electricity generation, water

supply for domestic and industrial use, flood control, recreation, inland navigation and fish breeding. **Jawaharlal Nehru** proudly proclaimed dams as the "**temples of modern India**", because they would integrate development of agriculture and village economy with rapid industrialisation and urban growth.

Examples: In the **Sutlej-Beas basin**, the **Bhakra Nangal** project water is used both for **hydel power and irrigation**. The **Hirakud** project in the **Mahanadi** basin integrates **conservation of water with flood control**.

Why multi-purpose projects are criticised:

1. **Regulating and damming rivers** affects their natural flow, causing poor sediment flow and excessive sedimentation at the bottom of the reservoir, resulting in rockier stream beds and poorer habitats for aquatic life.
2. Dams **fragment rivers**, making it difficult for aquatic fauna to migrate, especially for spawning.
3. Reservoirs on floodplains **submerge existing vegetation and soil**, leading to decomposition.
4. **Large-scale displacement** of local communities, who often do not get the benefits; landless and poor suffer while big landowners and industrialists benefit.
5. **Irrigation has changed cropping patterns** toward water-intensive commercial crops, causing salinisation of soil.
6. Dams did not always control floods; sedimentation reduced flood-carrying capacity, and **sudden release of water** during heavy rain worsened floods (e.g., Maharashtra and Gujarat in 2006).
7. They may induce **earthquakes**, water-borne diseases, pests and pollution from excessive use of water.

Movements against dams: the **Narmada Bachao Andolan** (Save Narmada Movement), a non-governmental organisation that mobilised tribal people, farmers, environmentalists and human rights activists against the **Sardar Sarovar Dam** on the Narmada in Gujarat. It started focusing on environmental issues

related to trees that would be submerged, and later shifted to rehabilitation of displaced people. The **Tehri Dam Andolan** is another example.

Inter-state water disputes have also arisen, e.g., the **Krishna-Godavari dispute**: objections raised by Karnataka and Andhra Pradesh to the diversion of more water at **Koyna** by the Maharashtra government for a multipurpose project, which would reduce downstream flow in their states.

Rainwater harvesting

Many thought that given the disadvantages and rising resistance against multi-purpose projects, **water harvesting** was a viable socio-economically and environmentally feasible alternative.

TRADITIONAL METHODS BY REGION

- **Hilly and mountainous regions (western Himalayas)**: people built diversion channels like the **'guls' or 'kuls'** for agriculture.
- **Rajasthan (drinking water)**: **rooftop rainwater harvesting** was commonly practised.
- **Flood plains of Bengal**: people developed **inundation channels** to irrigate their fields.
- **Arid and semi-arid regions (Rajasthan)**: agricultural fields were converted into rain-fed storage structures that allowed water to stand and moisten the soil, like the **'khadins' in Jaisalmer** and **'johads'** in other parts.
- **Bikaner, Phalodi and Barmer (Rajasthan)**: almost all houses traditionally had underground tanks or **'tankas'** for storing drinking water, connected to sloping roofs by pipes. Rain falling on rooftops travelled down the pipes into the tanka. The first spell of rain was usually not collected, to clean the roofs. The rainwater, called **'palar pani'**, is considered the **purest form of natural water**. Tankas also keep rooms cool in summer.
- **Meghalaya**: a **200-year-old system of bamboo drip irrigation**. Water from streams and springs is diverted through bamboo pipes; **18–20**

litres of water enter the system per minute, and it is reduced to **20–80 drops per minute** at the site of the plant (used for black pepper and betel leaf).

Modern examples:

- **Gendathur**, a remote backward village in **Mysuru, Karnataka**: villagers installed household rooftop rainwater harvesting systems (about 200 households). The village earns the rare distinction of being rich in rainwater.
- **Shillong (Meghalaya)**: interestingly, although **Cherrapunji and Mawsynram**, 55 km away, receive the highest rainfall in the world, Shillong faces acute water shortage, so nearly every household has a rooftop rainwater harvesting structure.
- **Tamil Nadu** is the **first state in India** to make **rooftop rainwater harvesting compulsory** for all houses across the state; there are legal provisions to punish defaulters.
- In western Rajasthan, the practice of rooftop harvesting is declining because perennial **Rajasthan Canal (Indira Gandhi Canal)** water is now available, though some houses still keep tankas because they do not like the taste of tap water.

BHAKRA NANGAL VS HIRAKUD

Bhakra Nangal (Sutlej-Beas): hydel power + irrigation. **Hirakud** (Mahanadi): water conservation + flood control. Students often swap these.

BOARD FAVOURITES AND MAP WORK

"Why is water scarcity rising?", "Why are multi-purpose projects called temples of modern India, and why are they opposed?" (5 marks), "Describe rooftop rainwater harvesting in Rajasthan/Meghalaya", "Tamil Nadu and rooftop harvesting". **Map (dams):** locate **Salal (Chenab, J&K), Bhakra Nangal (Sutlej, Punjab–Himachal), Tehri (Bhagirathi, Uttarakhand), Rana Pratap Sagar (Chambal, Rajasthan), Sardar Sarovar (Narmada, Gujarat),**

Hirakud (Mahanadi, Odisha), Nagarjuna Sagar (Krishna, Telangana–Andhra Pradesh), Tungabhadra (Karnataka).

Quick recap

- 96.5% of water in oceans, 2.5% freshwater (about 70% of it as ice).
- Scarcity arises from population, irrigation, urbanisation, industry, pollution and unequal access.
- Multi-purpose projects: Nehru's "temples of modern India"; Bhakra Nangal (power + irrigation), Hirakud (conservation + flood control).
- Opposition: displacement, ecological damage, floods, salinisation; Narmada Bachao Andolan, Tehri.
- Traditional harvesting: guls/kuls, khadins, johads, tankas (palar pani), inundation channels, bamboo drip (Meghalaya).
- Tamil Nadu first state to make rooftop harvesting compulsory.

Practice questions

1. Who called dams the "temples of modern India"?
   (a) Mahatma Gandhi (b) Jawaharlal Nehru (c) Sardar Patel (d) B.R. Ambedkar
2. The Narmada Bachao Andolan opposed the construction of
   (a) the Tehri Dam (b) the Hirakud Dam (c) the Sardar Sarovar Dam (d) the Bhakra Nangal Dam
3. In Rajasthan, rainwater stored in tankas is called
   (a) kul (b) johad (c) khadin (d) palar pani
4. Which state was the first to make rooftop rainwater harvesting compulsory for all houses?
   (a) Tamil Nadu (b) Rajasthan (c) Meghalaya (d) Karnataka
5. The bamboo drip irrigation system is practised in
   (a) Gujarat (b) Kerala (c) Punjab (d) Meghalaya
6. The Hirakud project on the Mahanadi integrates conservation of water with
   (a) inland navigation (b) flood control (c) hydel power only (d) fish breeding only
7. Which percentage of the world's water is freshwater?
   (a) 96.5 per cent (b) 30 per cent (c) 2.5 per cent (d) 70 per cent
8. Explain any three reasons for water scarcity in India.
9. Why are multi-purpose river projects facing resistance? Give five reasons.

Answers: 1. b 2. c 3. d 4. a 5. d 6. b 7. c 8. see notes 9. see notes